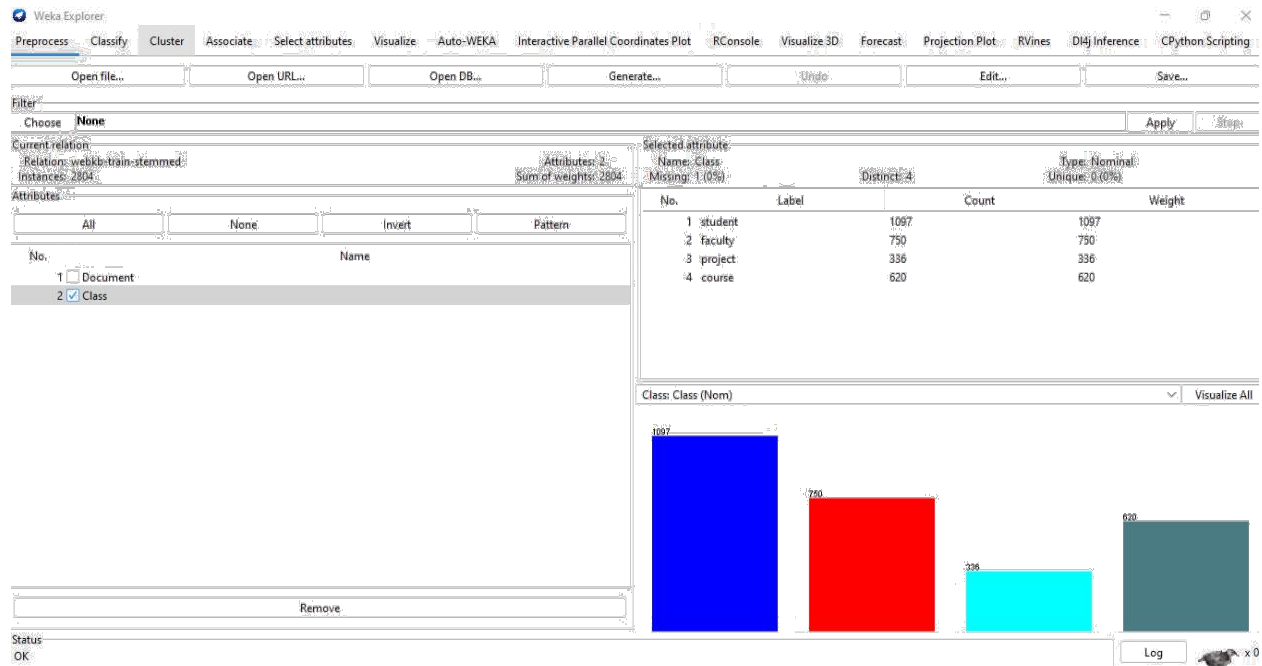


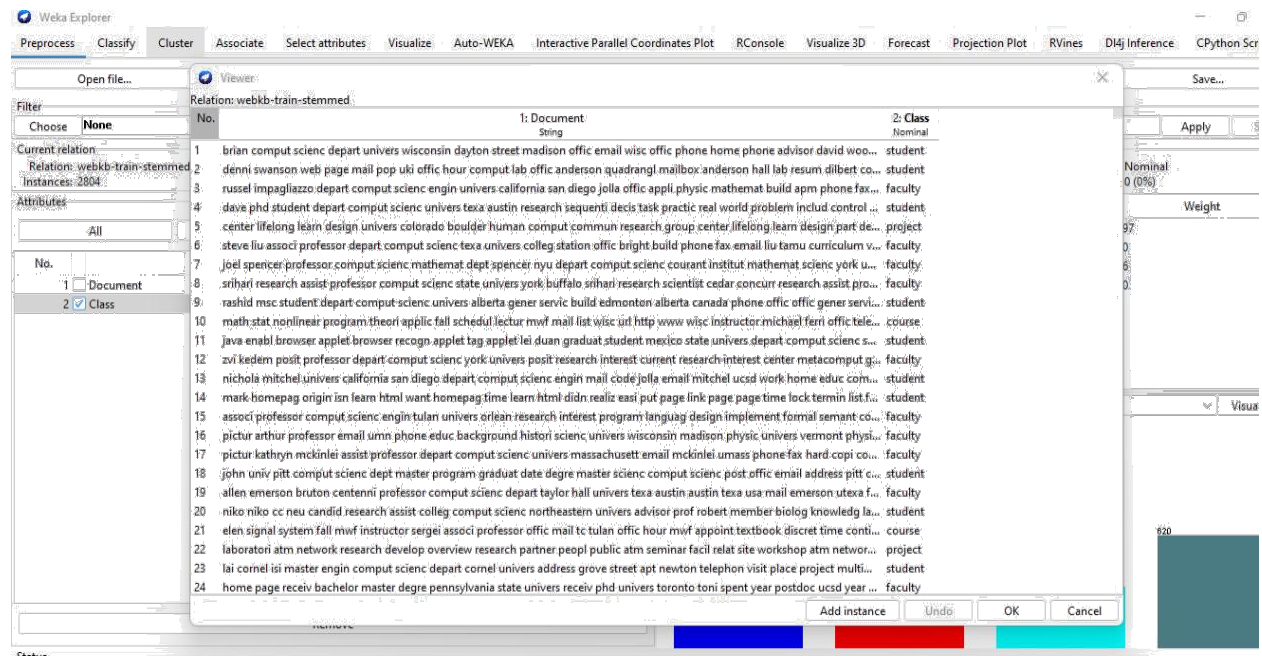
INFORMATION RETRIEVAL

THEJASWI SARUPURI

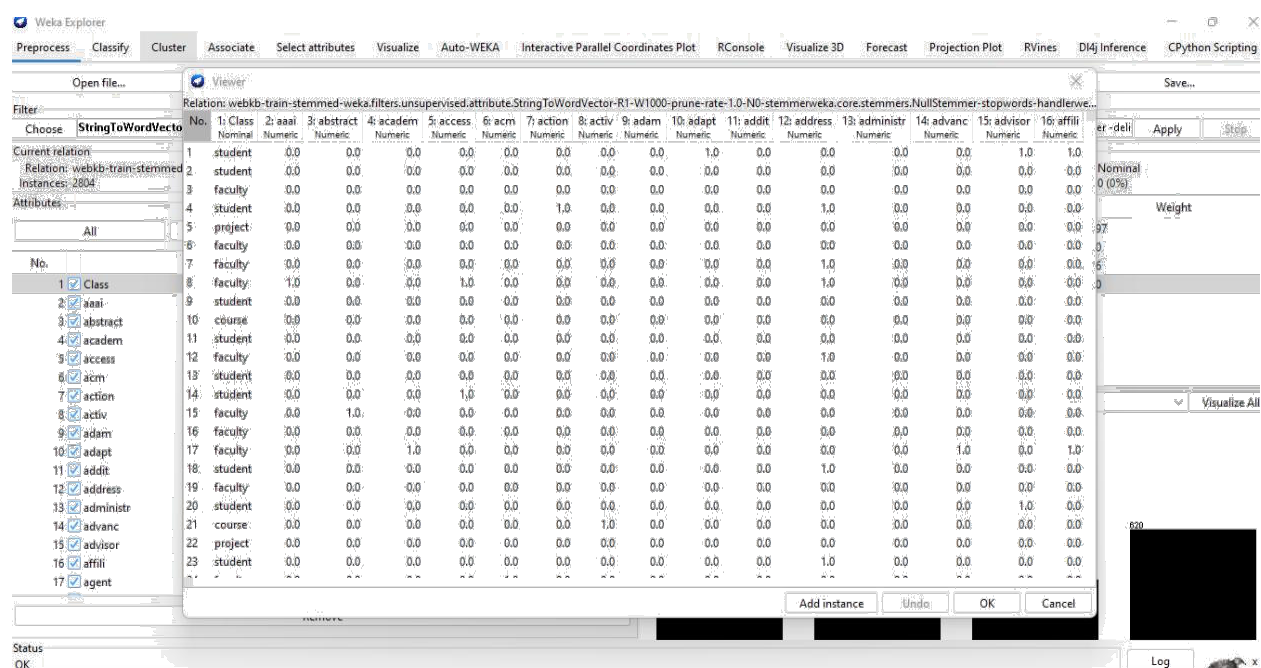
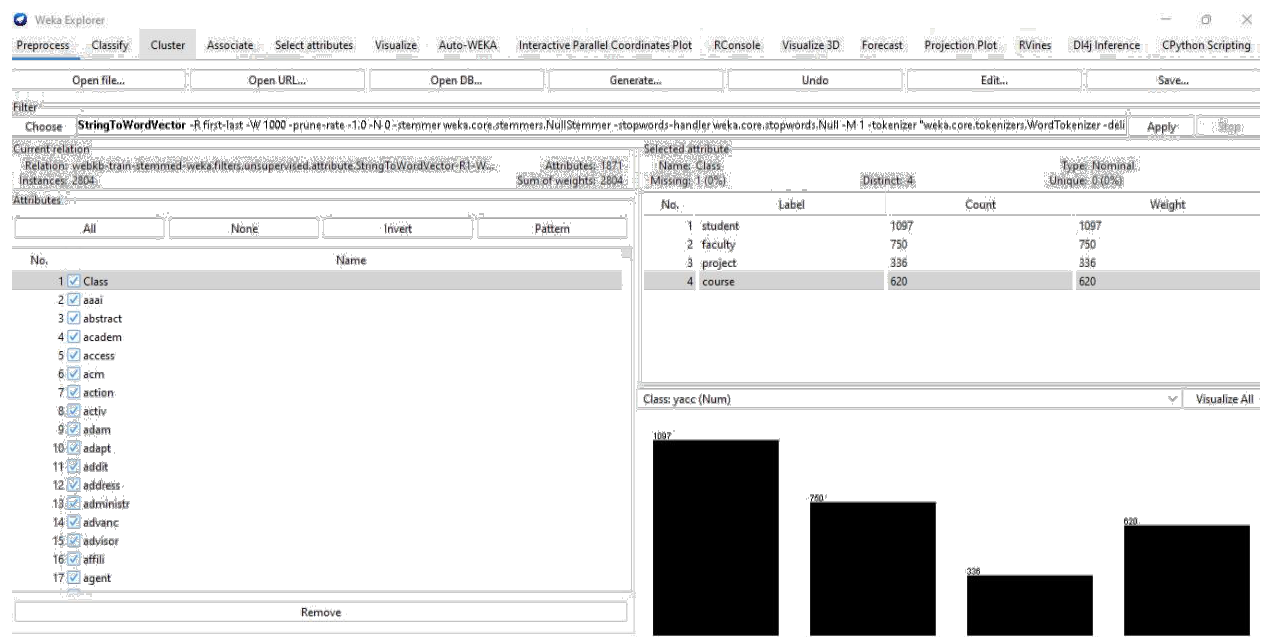
1. Load the Training Data set



One can examine and change the file after it has been loaded in the preprocess tab.



2. loading the training data set and creating the Document word matrix using the The Preprocess panel in Weka contains the "StringToWord" vector filter.



3. Conducting Classification using Naïve Bayes.

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Auto-WEKA

Interactive Parallel Coordinates Plot

RConsole

Visualize 3D

Forecast

Projection Plot

RVines

DI4j Inference

CPython Scripting

Classifier

Choose NaiveBayesMultinomialText -P 0 -M 3.0 -norm 1.0 -Inorm 2.0 -stopwords-handler weka.core.stopwords.Null -tokenizer "weka.core.tokenizers.WordTokenizer -delimiters |" -stemmer weka.core.stemmers.NullStemmer

Test options

☐ Use training set
 ☐ Supplied test set

Set...

☒ Cross-validation

Folds 10

☐ Percentage split

% 66

More options...

(Nom) Class

Start

Stop

Run on server

Result list (right-click for options)

16:58:02 - bayes.NaiveBayesMultinomialText

Classifier output

```

=== Run information ===

Scheme:      weka.classifiers.bayes.NaiveBayesMultinomialText -P 0 -M 3.0 -norm 1.0 -Inorm 2.0 -stopwords-handler weka.core.stopwords.Null -tokenizer "
Relation:     webkb-train-stemmed
Instances:    2804
Attributes:   2
              Document
              Class
Test mode:    10-fold cross-validation

=== Classifier model (full training set) ===

Dictionary size: 6770

The independent frequency of a class
-----
student 1088.0
faculty 751.0
project 837.0
course 621.0

The frequency of a word given the class
-----

```

student	faculty	project	course	
<laplace=1>	<laplace=1>	<laplace=1>	6.0	cancel
14.0	4.0	4.0	15.0	half
4.0	3.0	<laplace=1>	5.0	illumina
<laplace=1>	<laplace=1>	<laplace=1>	5.0	scalaroff
<laplace=1>	<laplace=1>	<laplace=1>	3.0	agupta
102.0	131.0	15.0	117.0	hall
4.0	<laplace=1>	<laplace=1>	<laplace=1>	clinton

Status

OK

Log

Weka Explorer

Preprocess **Classify** Cluster Associate Select attributes Visualize Auto-WEKA Interactive Parallel Coordinates Plot RConsole Visualize 3D Forecast Projection Plot RVines D4j inference CPython Scripting

Choose **NaiveBayesMultinomialText** - P 0 - M 3.0 - norm 1.0 - Inorm 2.0 - stopwords-handler weka.core.stopwords.Null - tokenizer "weka.core.tokenizers.WordTokenizer - delimiters "\n\t;\t;\t;\t;" - stemmer weka.core.stemmers.NullStemmer

Test options

☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds **10**
☐ Percentage split % **66**

[More options...](#)

Nom) Class

Start Stop Run on server

Result list (right-click for options)

16:58:02 - bayes.NaiveBayesMultinomialText

Classifier output

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      2400           85.6225 %
Incorrectly Classified Instances    403            14.3775 %
Kappa statistic                    0.7999
Mean absolute error                 0.0759
Root mean squared error             0.2517
Relative absolute error             21.3108 %
Root relative squared error         59.6497 %
Total Number of Instances          2803
Ignored Class Unknown Instances     1
```

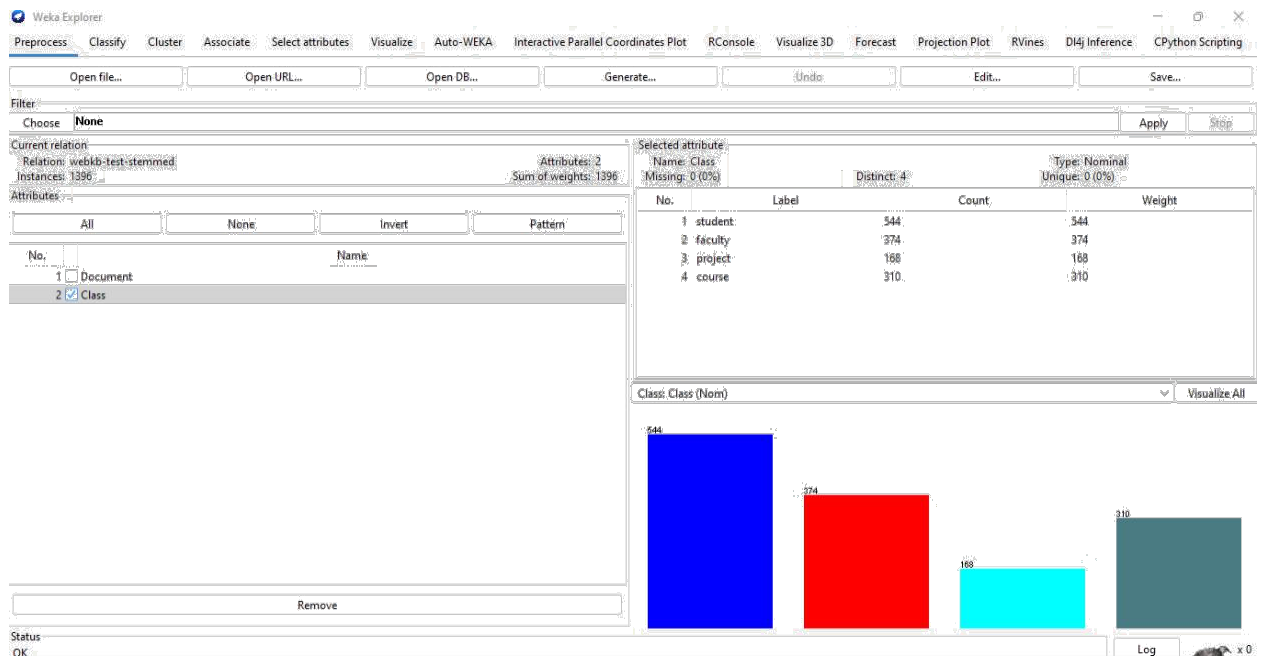
=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.855	0.064	0.896	0.855	0.875	0.798	0.956	0.935	student
	0.785	0.065	0.815	0.785	0.800	0.729	0.925	0.859	faculty
	0.854	0.053	0.688	0.854	0.762	0.732	0.963	0.765	project
	0.945	0.014	0.951	0.945	0.948	0.934	0.990	0.979	course
Weighted Avg.	0.856	0.052	0.862	0.856	0.858	0.802	0.956	0.904	

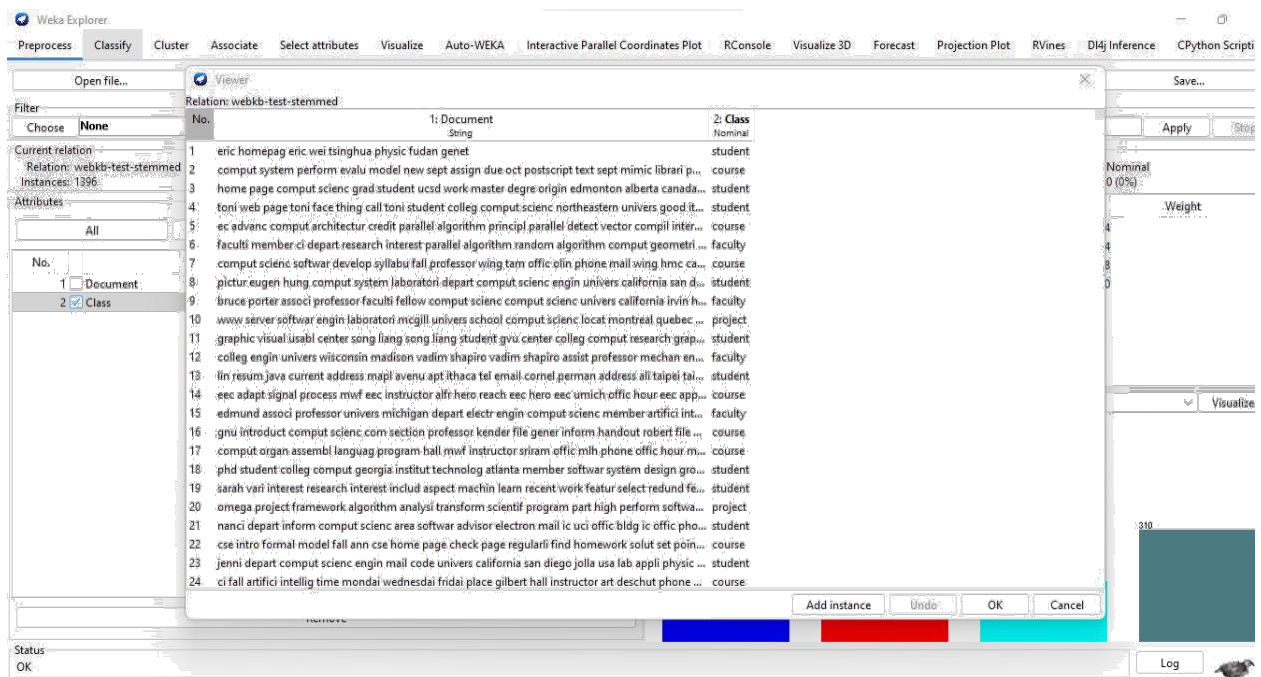
=== Confusion Matrix ===

```
a b c d <-- classified as
938 103 40 16 | a = student
84 589 69 8 | b = faculty
18 25 287 6 | c = project
7 6 21 586 | d = course
```

4. Loading the test dataset



One can examine and change the file after it has been loaded in the preprocess tab.



5. Generating the Document word matrix after loading the test data set by using the “StringToWord” vector filter in Weka in the Preprocess panel.

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize | Auto-WEKA | Interactive Parallel Coordinates Plot | RConsole | Visualize 3D | Forecast | Projection Plot | RVines | DI4j Inference | CPython Scripting

Open File... | Open URL... | Open DB... | Generate... | Undo | Edit... | Save...

Filter: Choose **StringToWordVector** -R first-last -W 1000 -prune-rate -1.0 -N 0 -stemmer weka.core.stemmers.NullStemmer -stopwords-handler weka.core.stopwords.Null -M 1 -tokenizer weka.core.tokenizers.WordTokenizer -delim Apply Stop

Current relation: Relation: webkb-test-stemmed-weka.filters.unsupervised.attribute.StringToWordVector-R1-W1... Attributes: 1942 Sum of weights: 1396

Attributes: All | None | Invert | Pattern

No.	Name
1	Class
2	aaai
3	abil
4	abstract
5	academ
6	accept
7	access
8	achiev
9	acl
10	acm
11	acquisit
12	action
13	activ
14	ad
15	adam
16	adapt
17	addit

Remove

Status: OK

Log

Selected attribute: Name: Class Missing: 0 (0%) Distinct: 4 Type: Nominal Unique: 0 (0%)

No.	Label	Count	Weight
1	student	544	544
2	faculty	374	374
3	project	168	168
4	course	310	310

Class: workload (Num) Visualize All

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize | Auto-WEKA | Interactive Parallel Coordinates Plot | RConsole | Visualize 3D | Forecast | Projection Plot | RVines | DI4j Inference | CPython Scripting

Open File... | Viewer | Save...

Filter: Choose **StringToWordVector** -R first-last -W 1000 -prune-rate -1.0 -N 0 -stemmer weka.core.stemmers.NullStemmer -stopwords-handler weka.core.stopwords.Null -M 1 -tokenizer weka.core.tokenizers.WordTokenizer -delim Apply Stop

Current relation: Relation: webkb-test-stemmed-weka.filters.unsupervised.attribute.StringToWordVector-R1-W1000-prune-rate-1.0-N0-stemmerweka.core.stemmers.NullStemmer-stopwords-handlerwe... Instances: 1396

Attributes: All | None | Invert | Pattern

No.	Class	aaai	abil	abstract	academ	accept	access	achiev	acl	acm	acquisit	action	activ	ad	adam	adapt	addit
1	student	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	course	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	student	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	student	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5	course	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6	faculty	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7	course	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8	student	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9	faculty	1.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
10	project	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
11	student	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
12	faculty	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
13	student	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
14	course	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
15	faculty	1.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
16	course	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
17	course	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
18	student	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
19	student	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
20	project	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
21	student	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
22	course	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
23	student	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Add instance | Undo | OK | Cancel

Status: OK

Log

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Auto-WEKA

Interactive Parallel Coordinates Plot

RConsole

Visualize 3D

Forecast

Projection Plot

RVines

Di4j Inference

CPython Scripting

Classifier

Choose

NaiveBayesMultinomialText -P 0 -M 3.0 -norm 1.0 -lnorm 2.0 -stopwords-handler weka.core.stopwords.Null -tokenizer "weka.core.tokenizers.WordTokenizer -delimiters '\n\r\t,;:\|'\"'0?!" -stemmer weka.core.stemmers.NullStemmer"

Test options

☐ Use training set

Set...

☒ Supplied test set

Folds 10

☐ Cross-validation

% 66

☐ Percentage split

More options...

(Nom) Class

Start

Stop

Run on server

Result list (right-click for options)

17:19:34 - bayes.NaiveBayesMultinomialText (Remote)

17:19:51 - bayes.NaiveBayesMultinomialText

Classifier output

=== Run information ===

Scheme:

weka.classifiers.bayes.NaiveBayesMultinomialText -P 0 -M 3.0 -norm 1.0 -lnorm 2.0 -stopwords-handler weka.core.stopwords.Null -tokenizer

Relation:

webkb-test-stemmed

Instances:

1396

Attributes:

Document

Class

Test mode:

user supplied test set: size unknown (reading incrementally)

=== Classifier model (full training set) ===

Dictionary size: 4554

The independent frequency of a class

student

545.0

faculty

375.0

project

169.0

course

311.0

The frequency of a word given the class

student

faculty

project

course

<laplace=1>

<laplace=1>

<laplace=1>

9.0

cancel

9.0

3.0

<laplace=1>

11.0

half

3.0

<laplace=1>

<laplace=1>

<laplace=1>

illumini

<laplace=1>

<laplace=1>

<laplace=1>

4.0

schieroff

58.0

79.0

5.0

45.0

hall

3.0

<laplace=1>

<laplace=1>

<laplace=1>

clinton

<laplace=1>

4.0

<laplace=1>

<laplace=1>

ksr

Status

OK

Log

 x 1

Weka Explorer

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Auto-WEKA

Interactive Parallel Coordinates Plot

RConsole

Visualize 3D

Forecast

Projection Plot

RVines

Di4j Inference

CPython Scripting

Classifier

Choose

NaiveBayesMultinomialText -P 0 -M 3.0 -norm 1.0 -lnorm 2.0 -stopwords-handler weka.core.stopwords.Null -tokenizer "weka.core.tokenizers.WordTokenizer -delimiters '\n\r\t,;:\|'\"'0?!" -stemmer weka.core.stemmers.NullStemmer"

Test options

☐ Use training set

Set...

☒ Supplied test set

Folds 10

☐ Cross-validation

% 66

☐ Percentage split

More options...

(Nom) Class

Start

Stop

Run on server

Result list (right-click for options)

17:19:34 - bayes.NaiveBayesMultinomialText (Remote)

17:19:51 - bayes.NaiveBayesMultinomialText

Classifier output

Time taken to test model on supplied test set: 0.51 seconds

=== Summary ===

Correctly Classified Instances

1294

92.6934 %

Incorrectly Classified Instances

102

7.3066 %

Kappa statistic

0.9978

Mean absolute error

0.0383

Root mean squared error

0.1717

Relative absolute error

10.7424 %

Root relative squared error

40.6762 %

Total Number of Instances

1396

=== Detailed Accuracy By Class ===

TP Rate

FP Rate

Precision

Recall

F-Measure

MCC

ROC Area

PRC Area

Class

0.921

0.045

0.929

0.921

0.925

0.878

0.966

0.962

student

0.901

0.023

0.934

0.901

0.917

0.808

0.962

0.966

faculty

0.929

0.024

0.939

0.929

0.981

0.866

0.992

0.929

project

0.968

0.009

0.966

0.968

0.968

0.959

0.996

0.993

course

Weighted Avg.

0.927

0.029

0.928

0.927

0.927

0.897

0.988

0.974

=== Confusion Matrix ===

a

b

c

d

<-- classified as

501

19

15

9

| a = student

24

337

13

0

| b = faculty

8

3

156

1

| c = project

6

2

2

300

| d = course

Status

OK

Log

 x 1